Blair Subbaraman

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Education

2020-Present PhD, Human Centered Design & Engineering, *University of Washington*, Seattle, WA
Advisor: Nadya Peek
Committee: Mako Hill, Daniela Rosner, Casey Reas

2022 MS, Human Centered Design & Engineering, *University of Washington*, Seattle, WA

2018 BA, Physics, *Pomona College*, Claremont, CA
Advisor: Dwight Whitaker
Minor: Mathematics

Publications

Refereed Conference and Journal Articles

2026 **Blair Subbaraman** and Nadya Peek. From Copy/Paste to Copying Pastes: Supporting Replication in an Online Digital Fabrication Community. In Proceedings of the 2026 ACM CHI Conference on Human Factors in Computing Systems (CHI '26)

2025 **Blair Subbaraman**, Nathaneal Bursch, and Nadya Peek. It's Not the Shape, It's the Settings: Tools for Exploring, Documenting, and Sharing Physical Fabrication Parameters in 3D Printing. In Proceedings of the 2025 ACM CHI Conference on Human Factors in Computing Systems (CHI '25) 🏅 Best Paper Honorable Mention (Top 5% of all accepted papers)

2024 **Blair Subbaraman**, Orlando de Lange, Sam Ferguson, and Nadya Peek. The Duckbot: A System for Automated Imaging and Manipulation of Duckweed. In Plos one 19, no. 1 (2024)

2023 **Blair Subbaraman** and Nadya Peek. 3D Printers Don't Fix Themselves: How Maintenance is Part of Digital Fabrication. In Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23).

2023 **Blair Subbaraman**, Shenna Shim, and Nadya Peek. Forking a Sketch: How the OpenProcessing Community Uses Remixing to Collect, Annotate, Tune, and Extend Creative Code. In Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23).

2022 **Blair Subbaraman** and Nadya Peek. P5.fab: Direct Control of Digital Fabrication Machines from a Creative Coding Environment. In Proceedings of the 2022 ACM Designing Interactive Systems Conference (DIS '22).

2022 Xiaowen Sun, Daniel Gilman Calderón, **Blair Subbaraman**, and Jeffrey A. Burke. An Audience Data-Driven Alternate Reality Storytelling Design. In International Conference on Human-Computer Interaction (C&C '22).

Conference Posters and Juried Demonstrations

2024 **Blair Subbaraman** and Nadya Peek. Playing the Print: MIDI-based Fabrication Interfaces to Explore and Document Material Behavior. In Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24).

2022 **Blair Subbaraman** and Nadya Peek. Demonstrating p5.fab: Direct Control of Digital Fabrication Machines From a Creative Coding Environment. In Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct).

2021 **Blair Subbaraman** and Nadya Peek. Demonstrating p5.fab: Direct Control of Digital Fabrication Machines From a Creative Coding Environment. In ACM Symposium on Computational Fabrication (SCF '21).

2021 Gabrielle Benabdallah and **Blair Subbaraman**. BodyTeller: Explorations in Narrative Biosensing, In Society for Social Studies of Science (4S '21).

Other Writing

2025 **Blair Subbaraman**. Printed Patterns. Algorithmic Pattern Conference (Alpaca 2025). Sheffield, UK. 2025.

2025 Brenden Pelkie et al. (including **Blair Subbaraman**). Democratizing Self-driving Labs Through User-developed Automation Infrastructure. ChemRxiv. 2025.

2022 Gabrielle Benabdallah and **Blair Subbaraman**. Explorations in narrative biosensing. Interactions. ACM New York, NY, USA, 2022.

Professional Activities

Talks

** Indicates invited talk*

2026 Tools and Community for Exploratory 3D Printing, *Teardown*, Portland, OR.

2026 Automation for Insight, *University of Bristol*, Bristol, UK.

2025 Automation for Insight, *HCDE Speaker Series*, Seattle, WA, USA.

2025 Printed Patterns, *Algorithmic Pattern Conference*, Sheffield, England.

2025 Automation for Insight, *NC State Building Future Faculty Program*, Raleigh, NC, USA.

2024 Sketching with Machines, *Hackaday Superconference*, Pasadena, CA, USA.

2024* Science Jubilee: Open-source Hardware for Laboratory Automation, *Duckweeds, Microbes, and Symbiosis*, University of British Columbia, Vancouver, Canada.

2023* Computational Fabrication in Performance Contexts, *UCLA School of Theater, Film, & Television*, Los Angeles, CA, USA.

2023 Bridging Creative Code and Digital Fabrication, *Open Source Arts Contributors Conference*, Clinic for Open Source Arts (COSA), University of Denver, CO, USA.

2023 Custom Workflow Automation with Jubilee, *Global Community Biosummit 6.0*, Virtual & MIT

2022* Niche Experiments with Jubilee: Duckweed Growth Assays as a Case Study, *Duckweeds and Microbes: A New Frontier in Symbiosis Research*, University of Toronto.

Workshops

2024-2026 Pathways to Open-Source Hardware for Laboratory Automation, *University of Washington*, Seattle, Washington, USA, Co-Organizer.
Co-organized three consecutive NSF-sponsored three-day workshop with >25 participants. The workshops gathered an international group of scientists and engineers from academia, industry, and national labs to build and share their approaches to open-source technologies for automating scientific experiments.

2025 3D Printing with p5.fab, *Creative Coding Fest*, Online
Organized multiple online workshops with over 20 educators and hobbyists to introduce 3D printing techniques using Javascript code.

2021 Creative Fabrication Workshop, *University of Washington*
Workshop with professional artists and seasoned makers to explore the possibilities of controlling machines with code.

Awards and Honors

2026 UW-RUA Research Exchange Award ($2,500)

2025 CHI Best Paper Honorable Mention (top 5% of publications)

2024 Processing Foundation Fellowship Finalist ($1,000)

2018 Richard P. Edmunds Physics Prize for outstanding graduating student, Pomona College

Teaching Experience

2025 Instructor of Record, HCDE 439: Physical Computing, *University of Washington*
Undergraduate level introduction to engineering and prototyping interactive systems & environments for human-centered applications. 40 students.

2021 Graduate Teaching Assistant, HCDE 439: Physical Computing, *University of Washington*
Sole TA, 25 students.

2018 Undergraduate Teaching Assistant, PHYS 128: Electronics with Laboratory, *Pomona College*
Transistors and integrated circuits in a variety of applications including operational amplifiers, basic digital circuits, analog/digital conversion and an introduction to microprocessors. 2 TAs, 30 students.

2017 Undergraduate Teaching Assistant, PHYS 41: General Physics with Laboratory, *Pomona College*
Calculus-based introduction to Newtonian mechanics and thermodynamics for non-majors. 2 TAs, 25 students.

2015, 2016 Undergraduate Teaching Assistant, PHYS 70: Spacetime, Quanta and Entropy with Laboratory, *Pomona College*
Calculus-based introduction to principles of contemporary physics. Topics include conservation laws, special relativity, quantum physics and thermal physics, all viewed from a 21st century perspective. 2 TAs, 30 students.

Mentorship

* *Signifies co-authorship on peer-reviewed article*

2025-Present Catherine Zhang, Bachelor's Student, Computer Science, University of Washington.

2024-2025 Nathaneal Bursch*, Bachelor's Student, Mechanical Engineering, University of Washington.

2022-2024 Sam Ferguson*, Bachelor's Student, Human Centered Design and Engineering, University of Washington.

2022-2023 Shenna Shim*, Bachelor's Student, Human Centered Design and Engineering, University of Washington.

2021-2022 Annie Liu, Bachelor's Student, Human Centered Design and Engineering, University of Washington.

2021-2022 Kenny Le, Bachelor's Student, Human Centered Design and Engineering, University of Washington.

Service

Reviewing

* *Special Recognition for Outstanding Reviews*

2026 ACM Conference on Human Factors in Computing (CHI)*
ACM Symposium on User Interface Software and Technology (UIST)*

2025 ACM Conference on Human Factors in Computing (CHI)*
ACM Conference on Designing Interactive Systems (DIS)*
ACM Symposium on User Interface Software and Technology (UIST)*

2024	ACM Conference on Human Factors in Computing (CHI)* ACM Conference on Designing Interactive Systems (DIS)* Halfway to the Future Symposium (HttF)
2023	ACM Conference on Human Factors in Computing (CHI)* ACM Conference on Designing Interactive Systems (DIS)*

Conference Service

2024	Student Volunteer, ACM Conference on Human Factors in Computing (CHI)
2022	Session Chair, ACM Symposium on Computational Fabrication (SCF) Student Volunteer, Eyeo Festival
2017	Student Volunteer, Eyeo Festival

Research Positions

2020-Present	Machine Agency with Nadya Peek, University of Washington Topics: Digital and computational fabrication, laboratory automation
2022	Slip Rabbit Studio with Timea Tihanyi & Audrey Desjardins, University of Washington Topics: Data physicalization, ceramic 3D printing
2021	Tactile and Tactical Design Lab with Daniela Rosner, University of Washington Topics: Biosensing, wearable technologies
2018-2020	Center for Research in Engineering, Media, & Performance (REMAP) with Jeff Burke, UCLA Topics: AR theater performance, open source software for the arts
2014-2018	Fluids Dynamics Lab with Dwight Whitaker, Pomona College Topics: Biological hydrodynamics, scientific instrumentation

Performances & Exhibitions

2020 The Invention of Morel, Virtual Performance
Developer for a workshop and performance exploring distributed remote virtual production during COVID-19 through an adaptation of The Invention of Morel by Adolfo Bioy Casares.

2019-2020 The Man in the High Castle, UCLA Freud Playhouse & Virtual
AR app development and HCI research for an experimental, immersive theater piece based on the world of the streaming series The Man in the High Castle.

2019 @LAs, UCLA Freud Playhouse
Developer and designer working with directors, cinematographers, and set designers to realize an AI/ML driven pop-up exhibition prototype as part of UCLA's Future Storytelling Summer Institute 2019.

2018 The Buzz, UCLA Campus
Created and deployed the sound platform for a Raspberry Pi based audio/furniture installation in collaboration with UCLA cityLAB.

2018 Entropy Bound, UCLA Little Theater
Developed software for custom wearables to support an experimental play weaving dramaturgy with code.

Selected Synergistic Activities

2021-Present p5.fab
Developed and maintain p5.js library for authoring physical artifacts.

2023-Present Science Jubilee
Developed and maintain open-source software and hardware for laboratory automation.

2018-2021 OpenPTrack Community Manager
Managed issues, questions, and discussions for OpenPTrack, an open source person tracking software for the arts.

2016-2018 Community Science Projects
Led partnership between Pomona College and Fremont Academy of Engineering & Design, a public high school in Pomona, CA.